B.Tech (ME)– IV-I Sem

23A03702b	**FINITE ELEMENT METHODS (Professional Elective-I)**	**L**	**T**	**P**	**C**
		3	**0**	**0**	**3**

	Course Objectives: The objectives of this course is to
1	Learn basic principles of finite element analysis procedure.
2	Gain knowledge of the concepts of Nodes and elements
3	Know the theory and characteristics of finite elements that represent engineering structures.
4	Apply finite element solutions to structural, thermal, and dynamic problems.
5	Develop the knowledge and skills to evaluate finite element analyses and apply design analysis Effectively.

COURSE OUTCOMES On successful completion of this course the students will be able to		
CO1	*Understand the concepts behind formulation methods in FEA*	L2, L3,L6
CO2	*Explain the concepts of Nodes and elements used in the analysis of beams and solve the simple problems*	L2, L4,L6
CO3	Understand the 2D stress analyses of the FEM method and solve the strain triangles.	L2,L4,L6
CO4	Apply Suitable boundary conditions to a global structural equation, and reduce it to a solvable form.	L1, L3,L5
CO5	Identify how the finite element method expands beyond the structural domain, for problems involving dynamics, heat transfer and fluid flow.	L1, L3,L5

UNIT I

Introduction to finite element methods

Introduction to finite element methods for solving field problems, applications, Stress and equilibrium, Boundary conditions, Strain-Displacement relations, Stress- strain relations for 2D and 3D Elastic problems. Potential energy and equilibrium, Rayleigh-Ritz method, Formulation of Finite Element Equations.

One dimensional Problems: Finite element modelling of ID bar elements coordinates and shape functions. Requirements for Convergence and Interpolation functions, Pascal's Triangle, Assembly of global stiffness matrix and load vector. Finite element equations, Treatment of boundary conditions, Quadratic shape functions.

UNIT II

1D Analysis of Trusses and Beams

Analysis of trusses: Stiffness Matrix for 1D truss element, Stress Calculations and Problems with maximum of three elements.

Analysis of beams: Element Stiffness Matrix and Load vector for 1 D beam element, Hermite shape functions and simple problems.

UNIT III

2D Analysis

Finite element modeling of two dimensional stress analysis with constant strain triangles and treatment of boundary conditions. Estimation of load Vector, Stresses.

Finite element modeling of Axi-symmetric solids subjected to axi-symmetric loading with triangular elements.

UNIT IV

Quadrilateral Elements & Thermal Analysis

Quadrilateral Elements: Isoparametric, Sub parametric and Super parametric elements, Modelling of 4 noded and 8 noded quadrilateral elements and simple problems. Numerical Integration.
Steady state heat transfer analysis: One dimensional analysis of composite slab and fin.

UNITV
Dynamic analysis
Analysis of a 1D uniform shaft subjected to torsion – Simple problems
Dynamic analysis: Formulation of finite element model, element – mass matrices, evaluation of Eigen values and Eigen vectors for a bar and shaft.

Textbooks:

1. T. Chandraputla, Ashok Belegundu, Introduction to Finite Element in Engineering, Pearson Publication 2011.
2. S.S.Rao, The Finite Element Methods in Engineering, Elsevier Butterworth -Heinemann, 2/e, 2011.
3. J N Reddy, An introduction to the Finite Element Method, McGraw – Hill, New York, 1993.

Reference Books:

1. R D Cook, D S Malkus and M E Plesha, Concepts and Applications of Finite Element Analysis, 3/e, John Wiley, New York, 1989.
2. K J Bathe, Finite Element Procedures in Engineering Analysis, Prentice-Hall, Englewood Cliffs, 1982.
3. G.LakshmiNarasaiah, Finite Element Analysis, 1/e, B.S. Publications, 2008.
4. O C Zienkiewicz and R L Taylor, the Finite Element Method, 3/e. McGraw-Hill, 1989.

Online Learning Resources:

https://nptel.ac.in/courses/112/104/112104193/
https://nptel.ac.in/courses/112/104/112104205/
https://nptel.ac.in/courses/105/105/105105041/
https://nptel.ac.in/courses/112/106/112106130/
https://nptel.ac.in/courses/112/103/112103295/